

AKIG TRANSACTION ANALYSIS - PROFESSIONAL PROJECT REPORT

1. Introduction

AKIG Transaction Analysis is a Business Intelligence (BI) solution developed to analyze banking transaction data and convert large volumes of raw records into meaningful business insights. Modern financial organizations generate thousands of customer transactions every day. Without a centralized analytics platform, it becomes difficult to identify spending trends, monitor customer behavior, evaluate card usage, and make informed business decisions.

The objective of this project is to design an interactive dashboard that provides a complete overview of banking transactions. Data is stored in MySQL, cleaned and transformed using Python in Jupyter Notebook, and finally visualized in Power BI. The dashboard enables users to monitor KPIs, filter information dynamically, and analyze customer activity from different business perspectives.

2. Project Objectives

The primary objective of this project is to build a secure, scalable, and interactive reporting solution. Specific objectives include:

- Store structured banking data in MySQL.
- Clean missing, duplicate, and inconsistent records using Python.
- Create a reliable data model in Power BI.
- Generate KPIs such as Total Customers, Total Transactions, Total Cards, Total Amount, and Highest Transaction.
- Provide filters for Card Brand, Gender, and Card Type.
- Improve business decision-making through visual analytics.

3. Technology Stack

MySQL was used as the relational database for storing customer, card, and transaction information. It provides efficient querying and data integrity.

Jupyter Notebook was used for preprocessing using Python libraries such as Pandas and NumPy. Data cleaning included removing duplicates, handling null values, converting data types, validating records, and preparing analytical datasets.

Power BI was used to connect with MySQL, create relationships, define DAX measures, build KPIs, and develop an interactive dashboard with slicers, charts, and drill-down analysis.

4. System Workflow

- Step 1: Banking transaction datasets are imported into MySQL.
- Step 2: Data quality checks are performed in Jupyter Notebook.
- Step 3: Clean datasets are loaded into Power BI.
- Step 4: Relationships are created between Customer, Transaction, and Card tables.
- Step 5: DAX measures calculate business KPIs.
- Step 6: Interactive charts and filters are developed.
- Step 7: Users analyze trends and make business decisions using the dashboard.

5. Dashboard Description

The dashboard contains multiple business visualizations.

KPI Cards:

- Total Customers – Displays the total customer base.
- Total Transactions – Shows the total number of processed transactions.
- Total Cards – Indicates active banking cards.
- Total Transaction Amount – Displays cumulative transaction value.
- Highest Transaction – Highlights the maximum transaction amount.

Charts:

- Transaction Over Time identifies monthly trends.
- Merchant City Analysis compares transaction amounts across cities.
- Transaction Type Analysis compares Credit, Debit, and Prepaid usage.
- Top Customers identifies high-value customers.
- Gender Analysis compares male and female customer distribution.
- Age Group Analysis helps understand customer demographics.

Interactive slicers allow users to filter reports by Card Brand, Gender, and Card Type.

6. Business Benefits

The solution helps financial institutions monitor operational performance and customer behavior. Decision-makers can identify high-value customers, monitor growth, compare merchant performance, evaluate customer demographics, and optimize marketing strategies. The dashboard significantly reduces manual reporting effort and provides faster access to reliable business insights.

7. Challenges and Solutions

Challenges included inconsistent transaction records, missing values, duplicate customer information, and integrating multiple datasets. These issues were resolved through Python-based preprocessing, SQL validation, and Power BI data modeling. Performance optimization techniques such as efficient relationships and calculated measures improved dashboard responsiveness.

8. My Contribution

I was responsible for designing the complete reporting workflow. My work included database understanding, data cleaning in Jupyter Notebook, connecting MySQL with Power BI, creating relationships, developing DAX measures, designing KPIs, implementing interactive visualizations, validating outputs, and testing dashboard performance. I also ensured that business users could easily navigate the dashboard using filters and slicers.

9. Future Scope

Future enhancements include real-time database connectivity, machine learning-based fraud detection, predictive customer analytics, automated report refresh, mobile dashboard optimization, and role-based access control. These improvements will further enhance scalability and business value.

10. Conclusion

AKIG Transaction Analysis successfully demonstrates how Business Intelligence technologies can transform raw banking transaction data into actionable insights. The integration of MySQL, Python, and Power BI provides an end-to-end analytical solution capable of supporting operational monitoring and strategic business decisions. The dashboard is scalable, interactive, and can be extended to support advanced analytics in future banking applications.